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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Bababoi
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8 / SDG 9
Project / Solution Name	WareOrbs
Name (Leader)	Alfred Lee Khai Zher
Email & Phone (Leader)	alfredleekz@gmail.com / 0175657674
University Affiliation(s)	Taylor's University
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small furniture retailers run warehouses that are full but hard to read. Stock is spread across floors, bays and pallets. The record of what arrived, what is reserved and what is free to sell lives in paper delivery notes, WhatsApp chats and the owner's memory.

This creates three costly gaps. Receiving: handwritten supplier notes are re-typed by hand, so counts and locations drift from reality. Slow stock: sofas and wardrobes that haven't sold for months keep occupying good bays. Nobody flags them, so sales staff never push them and the space is never freed for better-selling lines. Follow-through: when a buyer is found, staff repeat cargo sizes to several drivers, compare inconsistent delivery quotes and chase the owner for approval over chat, with no record of who decided what.

Three people are affected:

- The owner buys and prices on guesswork.
- Sales staff lack a clear list of what to sell.
- Warehouse workers search for items and reconcile errors by hand.

The economic cost is working capital tied up in ageing stock, wasted space and staff hours, and over-ordering the wrong products while bestsellers run out. Furniture is bulky, so every idle bay costs money. These SMEs can't afford enterprise warehouse systems, which is why the problem persists. Solving it supports SDG 8 (productivity and growth for small enterprises) and SDG 9 (giving SMEs access to digital infrastructure and innovation).

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

WareOrb is a web-based warehouse assistant that turns a furniture SME's stock records into daily action. The whole warehouse is shown as an interactive 3D map, and an AI agent called Jarvis guides each step.

How AI solves the problem:

1. **Smart receiving:** staff photograph a handwritten delivery note or speak the delivery. A large language model reads it, matches each line to the catalogue and builds a draft receipt. A worker confirms the real count and bay before stock is posted. AI never changes stock by itself.
2. **Ageing-stock alerts:** the system flags batches that have sat too long and explains why (age, quantity, bays occupied). It highlights them on the map and hands a ready-made product brief to a salesperson.
3. **Delivery quoting:** once a buyer is recorded, WareOrb combines the stored cargo size and weight with the buyer's address and requests delivery quotes through the Lalamove API. Incomplete or test quotes cannot be approved.
4. **Owner approval:** the owner sees sale price, delivery cost and net recovery on one card. Approval reserves the stock. Only the worker's "goods collected" confirmation reduces it.
5. **Replenishment advice:** from confirmed sales and incoming orders, the agent suggests what to reorder and how many, and the owner decides.

Jarvis works by voice (English and Mandarin) or typed commands. It finds products, flies the map to them and opens each workflow step. It can only use the tools that the user's role (owner, sales or warehouse) is allowed.

Skills from the SDC Phase 1 workshops we apply:

- **GenAI APIs:** structured document extraction and a tool-calling agent.
- **Database architecture:** a real-time ledger of SKUs, batches, reservations, movements and an audit log, with atomic writes and role-based access.
- **Cloud hosting:** a serverless backend and an HTTPS-deployed frontend with managed sign-in.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

- **Development:** Replit workspace with the Claude Opus 5.5 coding agent; code on GitHub
- **Frontend:** TypeScript, React 19, Bun, React Three Fiber + drei (Three.js) for the 3D warehouse, Tailwind CSS v4 + shadcn/ui, zustand
- **Backend & database:** Convex, which provides a real-time database, serverless functions, file storage for receipt photos, a scheduler and rate limiting
- **Authentication:** Clerk sign-in, with owner / sales / warehouse roles checked on the server
- **AI:** Anthropic Claude API for receipt reading (structured output), the Jarvis tool-calling agent and product icons.
- **Video generation:** Higgsfield's video-generation platform, connected to Replit's Claude Opus 5.5 agent through the Higgsfield MCP server
- **Logistics:** Lalamove API v3 for signed delivery quotes and vehicle sizes
- **Hosting & CI:** Vercel (HTTPS frontend), a Convex production deployment, and GitHub Actions running tests, type checks and the build

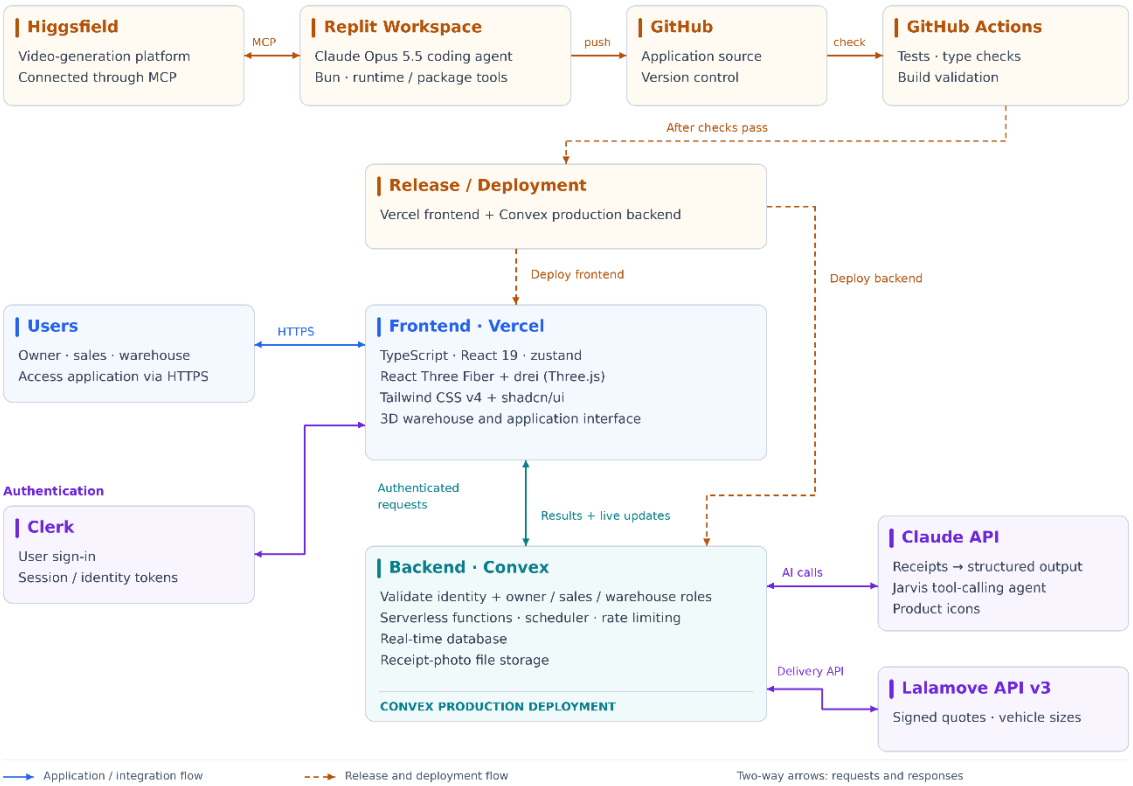
System diagram:

- Users sign in through Clerk and use the React and 3D frontend on Vercel.
- The frontend subscribes live to Convex, which checks roles and writes atomically to the database.
- Convex actions call the Claude, Lalamove and Fish Audio APIs.
- The app is built in Replit with the Opus 5.5 agent, which also drives Higgsfield video generation through MCP.

TECHNOLOGY STACK & SYSTEM FLOW

From Development to Warehouse Operations

Connected architecture based on the supplied technology stack



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: we mapped the daily work of the three people who touch stock: the owner, the salesperson and the warehouse worker. We found the pain was not missing records but records that never turned into action.

Define: give the business one trustworthy answer to "what is here, where is it, and what should we do about it?"

Ideate: we compared a narrow space-reallocation tool with a full operating loop and chose the loop: receive, flag ageing stock, sell, quote delivery, approve, dispatch, replenish. A human confirms every step that changes stock or money.

Prototype: built in Replit with Claude Opus 5.5. It has a 3D warehouse map staff instantly recognise, Jarvis voice and typed commands, and a separate view for each role.

Test: we scripted an end-to-end run across all three roles. We also wrote automated tests for unauthorised access, double posting, stale edits and competing reservations.

Deployment: the app runs on Vercel with a dedicated Convex production database and Clerk sign-in, starting from empty real business data. Owners invite staff by verified email. Every AI, voice and delivery feature has a manual fallback, so the shop keeps working if an API or network fails.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary: SDG Target 8.2.** *"Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors."* • WareOrb gives a labour-heavy furniture SME a technology upgrade. • AI reads delivery notes, flags ageing stock and prepares delivery quotes, so the same staff handle more stock with fewer errors. **Supporting: SDG Target 9.3.** *"Increase the access of small-scale industrial and other enterprises, in particular in developing countries, to financial services, including affordable credit, and their integration into value chains and markets."* • WareOrb connects an SME's idle stock back into markets. • Ageing goods are passed to salespeople, matched to buyers and shipped through Lalamove's logistics network, instead of sitting in the warehouse.

Metric Category	Your Response
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	• **Only 14% of Malaysian SMEs use inventory-management software** (Khazanah Research Institute, reported by TechWire Asia, 2021). Most still track stock on paper and in chat. • **65% of inventory records did not match physical stock** in a study of nearly 370,000 records across 37 stores (DeHoratius & Raman, 2008). This is our baseline for record accuracy: **35% of records correct.** • **Team estimate, to be confirmed by timed trial:** manually re-typing and allocating one 10-line supplier delivery note takes about 15 staff minutes.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. **Inventory record accuracy (main KPI).** Unit: % of SKU-and-bay records that match the physical count. 2. **Receiving productivity.** Unit: staff minutes per delivery note, from arrival to posted stock. How we measure: a test set of 20 sample delivery notes (handwritten and printed) with planted errors such as wrong SKU, wrong quantity and duplicate notes. Each note is processed once manually and once in WareOrb.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	• **≥95% inventory record accuracy** in the prototype test, up from the 35% published baseline. • **≥50% reduction in staff minutes per delivery note** compared with manual entry. This is about 15 → ≤7.5 minutes.

Metric Category	Your Response
	Both targets are realistic because WareOrb blocks stock from posting until a human confirms the count and bay, and it rejects duplicate notes automatically.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☒ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☒ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☒ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☒ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✓ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ✓ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✓ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

ALFRED LEE

24/10/2026